



Detrending temporal data affects stability drivers but not productivity stability in a species-rich grassland

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Abstract

1. Understanding the drivers of ecosystem stability generally requires disentangling how year-to-year fluctuations in species composition contribute to stability, particularly through species (a)synchrony and the temporal constancy of individual species. These metrics are often based on variance-derived quantities, such as coefficients of variation (CV) or log-variance ratios, and are typically calculated without considering the temporal order of observations. However, species composition and plant biomass may exhibit long-term trends driven by succession or persistent environmental change. Ignoring these trends can lead to misinterpretations of stability mechanisms.
 2. We propose and evaluate a suite of common indices broadly characterising stability and its drivers while accounting for long-term directional change in species composition. Our approach uses a two-term local quadrat variance measure (“TTLQV”) that isolates year-to-year fluctuations from long-term trends. Additionally, we introduce a new multivariate analogue of the coefficient of variation for compositional data. We show these methods using a 16-year dataset from a species-rich grassland, biomass-sorted samples collected under factorial treatments of fertilisation and dominant-species removal.
 3. Detrending did not affect the overall productivity stability but revealed changes in stability drivers based on community composition, including decreased asynchrony, and increased the stability of dominants. While both fertilisation and dominant removal consistently destabilised communities, the effects were rather independent of applying the detrending method. Asynchrony was found to be negatively correlated with biomass stability, likely due to the overriding destabilising fertilisation effects, which reduced species diversity and averaging effects.
- Synthesis: Failing to detrend time series can lead to misinterpretations of ecological stability drivers, such as possible overestimation of compensatory dynamics by asynchrony driven by species turnover and an underestimation of the stabilising contributions of dominant species.

Keywords: asynchrony; biodiversity; community dynamics; fertilisation; local variance; plant biomass; temporal variability

1 | INTRODUCTION

In a time of climate change and increasing climatic extremes, the stability of plant communities has become a crucial component for conserving biodiversity, maintaining biomass production, and sustaining ecosystem functioning. Defined broadly as “...the ability of a system to continue functioning under changing conditions or disturbances” (Pimm, 1984), ecological stability is a central concept in community ecology. However, its measurement and ecological context remain debated among scholars. One reason for this is that the interpretation strongly depends on how different facets of stability, for example, temporal variability, are quantified.

Every plant community exists in a dynamic state and undergoes constant changes driven by both internal and external factors (Jackson & Blois, 2015). The extent of these changes can range from subtle and barely noticeable to dramatic transformations in the community and can differently affect various components of community structure, like total biomass, species composition, species richness, etc. Internal factors, such as natural succession, population dynamics, and interactions, arise from processes within the community itself. In contrast, external factors originate from the environment outside the community, which includes various stressors and disturbance events, including different management regimes, that can induce shifts in species composition, abundance and overall structure (Wang et al., 2018; Ravnjak et al., 2022; Hefty et al., 2025).

Several effects and mechanisms can contribute to the stability of plant productivity (Figure 1), including the dominance effect (stabilisation due to dominant species), species asynchrony (stabilisation due to compensations between species), and the averaging effect (Loreau & de Mazancourt, 2008; Lisner et al., 2024; Segrestin et al., 2024; Lepš & Lisner, 2025). Plant community stability is typically quantified by examining how consistently communities maintain their structure and function over time. It is widely accepted that dominant species can stabilise communities when their own populations are stable (Gracia et al., 2026). In long-term studies, one of the most analysed facets is temporal constancy (invariability), the magnitude of temporal variability around a “steady state”. This magnitude is usually quantified using measures based on the coefficient of variation (CV). The most commonly used state variables to describe aspects of community stability are species composition, total biomass and population-level properties (Figure 1). The stability of total biomass is often emphasised because of its connection to productivity and ecosystem functioning (Hautier et al., 2014; Lisner et al., 2024). In grassland ecosystems, as in this study,

productivity stability can be approximated by the stability of vegetation peak biomass (Byrne et al., 2011). Other indices considered are also based on variance-derived quantities, such as log-variance ratios for synchrony between species.

An important aspect, often overlooked, when studying stability and its components, is that most of the metrics, including CV or log-variance ratios, are typically calculated without considering the temporal order of observations. By this, they fail to disentangle long-term directional trends from year-to-year variation (Lepš et al., 2019; Segrestin et al., 2024). In long-term datasets, these directional trends are common (caused by e.g. climate change or choice of management) and can conceal annual effects (Jackson & Blois, 2015; Lepš & Lisner, 2025). Previous studies have proposed to compute synchrony indices after taking into account directional trends (Lepš et al., 2019), showing greater synchrony between species (Valencia, Bello, et al., 2020). One approach to separate these mechanisms is to use the traditional indices (CV-based or log-variance for synchrony) alongside a “detrended” approach, e.g. two-term local quadrat variance (TTLQV) (Hill, 1973). Originally developed for spatial data, it is calculated from differences between neighbouring quadrats – or subsequent timepoints and when used for temporal data (Lepš et al., 2019), isolates year-to-year variation while removing the effect of temporal autocorrelation and linear trends (Hill, 1973; Lepš et al., 2019). It is important to note that autocorrelation does not necessarily imply a trend, but a trend inherently causes autocorrelation. Detrended and not detrended approaches do not represent a “more” or “less” correct way to quantify stability; they rather emphasise different temporal components, with detrending focusing on, for example, year-to-year fluctuation beyond trends.

Here, we expand the original idea of detrending stability drivers, initiated by Lepš et al. (2019) on synchrony indices, by proposing a suite of indices for characterising different stability components, including stability of productivity, composition, and population stability, providing a comprehensive tool that covers the main metrics considered when assessing ecological stability (Figure 1). After characterising the new indices, we use a 16-year dataset of a grassland experiment with a factorial design consisting of fertilisation and the removal of dominant species as treatments. We evaluate how year-to-year variation and long-term trends affect different facets of stability and apply classical CV-based measures, as well as their multivariate counterparts (Figure 1): temporal constancy of total biomass, population-level variability, and species composition, characterised by Euclidean and chord distance to address whether detrending alters the estimates of variability of biomass and species composition. Additionally, we investigate how detrending can alter the ecological interpretations of the

facets and drivers of stability. The use of manipulative experiments is particularly relevant because they isolate the possible effect of global change components. In this case, we deal with fertilisation and the removal of dominant species. Whereas fertilisation mimics the consequences of (mis)management and eutrophication, removal of the dominant species was introduced to the experiment as a manipulation, showing the role of dominant species in a community. Fertilisation can destabilise communities by shifting the physiological limitation of plants from nutrients to weather-driven variability (Lepš 2004), while simultaneously decreasing species richness and altering dominance structure (Hautier et al., 2014, 2020) and generally decreasing synchrony (Valencia, de Bello, et al., 2020). Removal of dominant species can also disrupt the stabilising effect on the community, as shown by Lepš et al. (2019), as well as Segrestin et al. (2024).

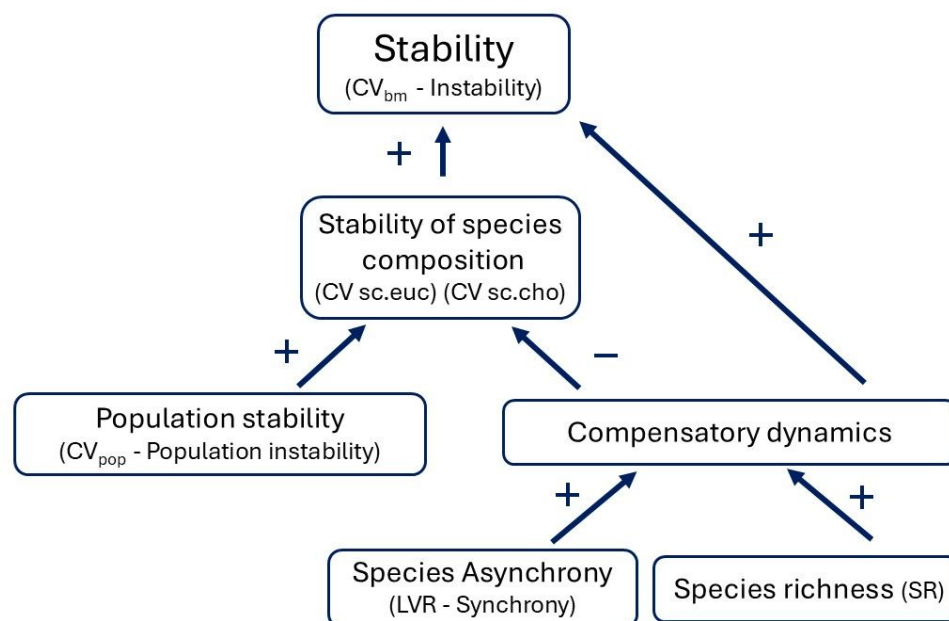


Figure 1: Conceptual framework of stability components. Conceptual model illustrating the pathways through which temporal stability is influenced by individual population stability and compensatory dynamics among species. Arrows represent hypothesised causal relationships between species richness, asynchrony, and compositional stability. Compositional stability is quantified using multivariate coefficients of variation (CV) based on Euclidean and chord distances.

2 | METHODS

2.1 | Study area and data collection

This study was conducted in the oligotrophic, species-rich meadow in Ohrazení, South Bohemia, Czech Republic (48°57'0" N, 14°36'0" E; 510 m above sea level), where a long-term experiment has been running since 1994. For this study, we used data from 1999 to 2014. Mean annual temperature increased from 8.9°C to 10.1°C between 1999 and 2014 ($\Delta = 1.2^\circ\text{C}$), and mean annual precipitation increased from 714.8 mm to 781.7 mm over the same period ($\Delta = 66.9$ mm). However, neither variable showed a significant trend over time. The minimum and maximum yearly average temperatures ranged from 7.9°C to 10.1°C, and precipitation ranged from 514.0 to 1009.8 mm (Figure S1 in Supplementary Material). Climate data were obtained from the Climate Change model provided by (Meteoblue. (2025). *Climate Data for Ohrazení, Czech Republic.*), which visualises how global warming has affected specific regions for the past eight decades. The model uses ERA5 data from the European Centre for Medium-Range Weather Forecasts (ECMWF).

The vegetation of this meadow belongs to *Molinion caeruleae* Koch 1926 alliance, with some elements of *Violion caninae* Schwickerath 1944 alliance. Dominant graminoids of this site include *Molinia caerulea* (L.) Moench, *Nardus stricta* L., *Festuca rubra* L., and *Holcus lanatus* L. The meadow also hosts several sedge species, such as *Carex panicea* L., *Carex hartmanii* Cajander, *Carex nigra* (L.) Reichard, *Carex pallescens* L., and *Carex pulicaris* L. The dominant forb species comprises *Potentilla erecta* (L.) Räusch., *Ranunculus acris* L., *Succisa pratensis* Moench, and *Stachys officinalis* (L.) Trevis.

The experiment consists of 24 permanently marked plots (2 × 2 m). The plots were arranged in a contiguous rectangular layout comprising six rows and four columns. In the first year when the experiment began, a factorial combination of fertilisation, mowing, and removal of the dominant species *Molinia caerulea* was applied, resulting in eight treatment combinations (each treatment combination was replicated three times) (Figure S2 in Supplementary Material). Before the establishment of the experiment, the grassland was regularly mowed and managed using traditional practices. However, this practice was stopped 3–4 years before the start of the experiment in 1994 (Lepš, 1999). Mowing was subsequently resumed in 1994 and carried out annually in the second half of June. Mowing was applied as clipping of biomass at approximately 1 – 2 cm within the central 50 x 50 cm quadrat and simultaneously served as the biomass sampling method; biomass data are not available for the

unmown plots (Lepš, 2004). The rest of the 2 m x 2 m plot was sickled at approximately the same height at the same time. At the beginning of the experiment (from 1994 to 1996), fertilisation was applied twice a year at a total rate of 65 g/m² (50 g/m² in autumn and 15 g/m² in spring). A commercial NPK fertiliser was used, containing 12% N (as nitrate and ammonium), 19% P (as P₂O₅), and 19% K (as K₂O). From 1997 to 2002, the same fertiliser and dosage were applied, but only once a year in spring. Since 2003, the commercial fertiliser has been replaced by Cererit (8% N, 13% P, 11% K), and the application rate has been reduced to 50 g/m² per year. From April 1995, *Molinia caerulea* tillers were individually manually removed using a screwdriver aiming to minimize the disturbance of soil and other species. Newly emerging individuals of *M. caerulea* were removed each year (if needed). Detailed descriptions and the rationale for the applied treatments are provided in (Lepš, 1999, 2014; Lepš & Lisner, 2025).

The initial sampling was conducted in 1994, before the implementation of the treatments. These data served as a baseline for evaluating treatment effects in subsequent years. Plant species composition and visually estimated each year in late May/early June in all the plots (at two spatial scales, 0.25 m² and 1 m²) – the data have been used in several previous studies (Lepš, 1999, 2004, 2014; Lepš et al., 2018, 2019; Lepš & Lisner, 2025) and served as important lead in biomass sorting. The first five years of the experiment (1994 – 1998) were governed by directional changes as a response to treatment introduction and were thus omitted from the recent analysis. The data from subsequent years were not included due to the change in management (Lepš & Lisner, 2025). Unmown plots were excluded because biomass data were unavailable.

In several plots, particularly in the species-rich unfertilized ones, up to 40 species may occur within a 0.25 m² area. In such diverse communities, accurate identification of many species, particularly grasses and sedges, becomes challenging after clipping, since the stem bases, which are crucial for identification, are not included in the harvested biomass. Therefore, individuals of taxa that are challenging to identify were identified directly in the field and immediately clipped and sorted (Fig. S3) – in our experience, it is the only possibility to get reliable composition in such species-rich vegetation. The remaining vegetation is then harvested and subsequently sorted into species in the laboratory.

2.2 | Stability Indices and Characterisation

In this study we focus on constancy as a primary facet of ecological stability. It is fundamental to recognise that the stability characteristics of a plant community are determined by the selected state variables, the specific measurements used to characterise the community. We consider the stability of total biomass and the stability of species composition to be the most critical indicators of community-level stability. In this study, productivity stability is quantified as the stability of peak vegetation biomass, a common proxy in temperate grasslands where spring biomass is negligible, and peak values effectively reflect annual productivity. While stability of total biomass is a central ecosystem function, the stability of species composition is often of primary interest from a conservation perspective, particularly in the context of endangered communities.

To quantify stability, we characterise its counterpart (variability) utilising the Coefficient of Variation ($CV_{bm} = \text{s.d.}/\text{mean}$) for univariate data and its multivariate analogues for compositional data. Note that under this framework, a higher value indicates lower constancy and, consequently, lower stability. The stability of biomass, whether for total community biomass or individual species, is calculated over time within each plot. The standard deviation (s.d.) is traditionally derived from the classical variance formula,

$$(1) \text{ var} = \frac{\sum_{i=1}^t (x_i - \text{mean})^2}{t - 1}$$

where t is the total length of the time series and x_i is the biomass in year i . As mentioned, classical variability characteristics are computed regardless of the chronological order of observations. As a result, a high variance may be the consequence of a long-term directional trend or temporal autocorrelation, even when year-to-year variation is relatively small. This distinction is ecologically significant because theories predicting temporal stability, such as those identifying the role of acquisitive traits (de Bello et al., 2021), frequently assume rapid, short-term reactions rather than long-term trends. To address this issue, we employ a detrending approach based on the two-term local quadrat variance ($TTLQV$), originally proposed by Hill (1973) for spatial pattern analysis and adapted here for temporal series (Lepš et al., 2019). This method emphasises year-to-year changes by focusing on the differences between subsequent values.

$$(2) \text{ TTLQV} = \frac{\sum_{i=1}^{t-1} (x_{i+1} - x_i)^2}{2 * (t - 1)}$$

While other detrending methods exist, such as the direct removal of linear trends or three-term local quadrat variance, we selected $TTLQV$ because its use of differences between

subsequent values intuitively corresponds to the emphasis on year-to-year fluctuations and aligns with classical compositional dissimilarity measures. If no autocorrelation or trend is present in the time series, both the classical variance and $TTLQV$ serve as estimates of the same underlying parameter. Consequently, we calculate stability using both the standard CV (e.g., CV_{bm}) and its detrended version ($ttCV_{bm}$), where the square root of $TTLQV$ is used in the numerator. This dual approach is applied to community biomass and the average of population-level variabilities (CV_{pop} and $ttCV_{pop}$). The population-level CV is calculated as a weighted average of individual species CV s, weighted by their average biomass, and represents the value the community CV would achieve if all species fluctuated in perfect synchrony (Segrestin et al., 2024).

The stability of species composition is assessed using multivariate analogies of the CV . Multivariate variance is defined as the average squared distance of annual observations from the centroid of the series:

$$(3) \text{ var}_{mv} = \frac{\sum_{i=1}^t (ED(SpCom_i - SpCom_{mean}))^2}{t - 1}$$

where $ED(SpCom_i - SpCom_{mean})$ is the Euclidean distance (ED) between the composition in year i and the average species composition. Similarly, we can also calculate the detrended analogy, i.e. the analogy of $TTLQV$.

$$(4) TTLQV_{mv} = \frac{\sum_{i=1}^{t-1} (ED(SpCom_i - SpCom_{i+1}))^2}{2 * (t - 1)}$$

As for the univariate case, here we also calculated the multivariate counterpart of s.d. as a square root of var_{mv} or $Ttlqv_{mv}$. To ensure comparability with univariate CV , this variance is scaled by the sample norm (SN), which is the length of the mean abundance vector

($SN = \sqrt{\sum_{i=1}^S x_i^2}$), where x_i is the average abundance of species i and S is total number of species (Pielou, 1984).

We utilise two distinct distance measures to capture different ecological signals. Euclidean distance is used to reflect fluctuations in abundances in absolute terms. However, because we are also interested in the variability of species proportions, we employ the chord distance which focuses on relative abundances to the total biomass of a community. Chord distance is defined as Euclidean distance, where values are standardised by the sample norm (Orlóci, 1967; Šmilauer & Lepš, 2014). This standardisation simplifies the calculation as the

denominator of the CV becomes unity, meaning the square root of the multivariate variance is not only analogous to the standard deviation but also directly represents the CV .

Finally, we quantify species synchrony to investigate the drivers of community stability. We use the log-variance ratio (LVR ; Schluter, 1984), which is relatively independent of species richness.

It is calculated as:

$$LVR = \log \left(\frac{\text{var}(\sum_{i=1}^S x_i)}{\sum_{i=1}^S \text{var}(x_i)} \right)$$

i.e. log of the ratio of the variance of total biomass and the sum of variances of individual species. Its value is zero if species fluctuate independently, positive under synchrony (concordant fluctuations), and negative under asynchrony (where a decrease in one species is compensated by an increase in another). Theoretically, asynchrony in individual species fluctuations, which necessarily implies increased variability in species proportions, should stabilise total community biomass. While we expect stability measures of dominant species to correlate positively with total biomass variability, the changes in relative proportions are mathematically independent of total biomass fluctuations. By using detrended versions of these indices, we ensure that our assessment of compensatory dynamics is not confounded by shared responses to long-term environmental trends.

2.3 | Data Analysis

The response of stability indices to experimental treatments was evaluated using a nested Analysis of Variance (ANOVA) framework to account for the hierarchical structure of the data in which, for each plot, we considered two values for each index of stability. For visual representation in boxplots (Figure 2 below), confidence intervals were calculated based on the raw data distribution to provide a descriptive overview of group-level variability. It is important to note that these visual markers are data-based rather than model-based; they do not incorporate the random error terms or the assumption of homogeneous variance shared across groups in our nested ANOVA framework. Consequently, statistical significance reflects the rigorous partitioning of variance within the hierarchical model, which may differ from the visual overlap of the raw confidence intervals. Moreover, fertilisation and removal treatments were treated as "between-plot" factors, while the indices themselves (standard vs. detrended)

and, where applicable, the standardisation methods (Euclidean vs. chord distance) were treated as "within-plot" factors. Thus, the "between-plot" factors reflect the experimental treatments, whereas the "within-plot" factors reflect the methodological choices. The experimental plots were included as a random error term to correctly partition the variance. Due to the limited sample size, our primary models were constructed without including interactions between the within-factor components, except in the case of compositional dissimilarity. For this specific metric, we considered the full interaction between standardisation and detrending to assess how different views of compositional change (absolute vs. relative) responded to temporal detrending.

To isolate the specific drivers of different stability types, we further conducted separate ANOVAs for classical and detrended indices. This approach allowed us to reserve sufficient degrees of freedom to test the main effects of treatments and the methodological effect of detrending, which served as a direct test of the influence of long-term trends versus year-to-year fluctuations. These detailed individual models are provided in the Supplementary Material.

Complementary to the univariate analyses, we performed a Principal Component Analysis (PCA) on correlation matrix to visually synthesise the relationships between the various stability facets. In the PCA on correlation matrix, all variables were centred and scaled to unit variance by default. Within this ordination space, the geometric properties of the biplot arrows are directly interpretable: the cosine of the angle between any two arrows reflects their Pearson correlation coefficient. Consequently, narrow angles indicate a high positive correlation, obtuse angles represent negative correlations, and perpendicular arrows suggest independence. (However, be aware that the reflection would be precise if the angles are measured in all the PCA dimensions – if shown just for the first and second PCA axis, it is just the best possible approximation.) This visualisation allowed us to verify the expected mathematical dependencies between total biomass variability and the fluctuations of dominant species, as well as the predicted independence of compositional stability calculated from relative species proportions. All univariate statistical analyses were performed in the R environment (R Core Team, 2023) using the *stats* package, with data manipulation and visualisations generated via the *dplyr* (Wickham, François, et al., 2023), *tidyr* (Wickham, Vaughan, et al., 2023), *vegan* (Oksanen et al., 2025) and *ggplot2* packages (Wickham, 2016). The multivariate analysis was conducted using the software Canoco5 (Ter Braak & Šmilauer, 2012).

3 | RESULTS

The experimental treatments and detrending application had a significant influence on various facets of community stability. Total biomass variability (CV_{bm}) was primarily driven by fertilisation, with fertilised plots (FE) exhibiting significantly higher variability, and thus lower stability, compared to unfertilised plots (NF) ($f_{1,8} = 7.46$, $p = 0.026$) (See Table 1, see Figure 2 for a comparison of CV_{bm} across treatments). Interestingly, detrending had a negligible effect on biomass stability ($f_{1,8} = 2.0$, $p = 0.195$), suggesting that the variability in total productivity is dominated by year-to-year fluctuations rather than long-term directional trends. This is further supported by the PCA, where CV_{bm} and its detrended counterpart ($ttCV_{bm}$) are positioned closely together, reflecting their high correlation (Pearson's $r = 0.76$, Figure 3). In contrast to total biomass, stability measures based on species composition and population dynamics were highly sensitive to both experimental treatment and detrending. Species composition variability (CV_{sc}) was significantly higher in plots where the dominant *Molinia caerulea* was removed (RE) compared to non-removal plots (NR) ($f_{1,8} = 14.53$, $p = 0.005$). The removal of the dominant species also led to higher average population variability (CV_{pop} ; $f_{1,8} = 12.3$, $p = 0.008$), suggesting that *Molinia caerulea* exerts a stabilising influence on the remaining community. Detrending had a marked effect on all compositional and population-level metrics, consistently producing lower values for species turnover and population variability ($ttCV_{sc}$: $f_{1,24} = 101$, $p < 0.001$; $ttCV_{pop}$: $f_{1,8} = 74$. PCA, in which CV_{bm} and its detrended counterpart ($tt CV_{bm}$) are closely aligned (< 0.001). This indicates that a substantial portion of the observed compositional change over the study period was due to long-term directional shifts rather than year-to-year-variability. Furthermore, the choice of standardisation (Euclidean vs. chord distance) significantly affected CV_{sc} estimates ($f_{1,24} = 60.15$, $p < 0.001$), reflecting different responses in absolute vs. relative species proportions. Regarding synchrony, the detrended log-variance ratio ($ttLVR$) was significantly higher than the standard LVR , revealing that removing long-term trends uncovers greater underlying year-to-year synchrony among species ($f_{1,8} = 6.154$, $p = 0.038$, Table 1).

To assess the consistency of treatment effects on detrended and non-detrended data, separate ANOVAs were performed (Table S1, Supplementary Material). Although the statistical strength of some drivers varied. Specifically, the impact of dominant species removal remained

highly significant across all population and compositional metrics in both classical and detrended frameworks (e.g., $CV_{sc.cho}: f_{1,8} = 15.05$, $p = 0.005$ vs. $ttCV_{sc.cho}: f_{1,8} = 11.73$, $p = 0.009$; $CV_{pop}: f_{1,8} = 14.07$, $p = 0.006$ vs. $ttCV_{pop}: f_{1,8} = 8.86$, $p = 0.018$, Table S1). In contrast, the effect of fertilisation appeared more sensitive to detrending; while it significantly increased biomass variability and synchrony in the classical indices ($CV_{bm}: f_{1,8} = 10.336$, $p = 0.012$; $LVR: f_{1,8} = 6.455$, $p = 0.035$), these effects were partially attenuated when considering only year-to-year fluctuations ($ttCV_{bm}: f_{1,8} = 3.970$, $p = 0.082$; $ttLVR: f_{1,8} = 0.701$, $p = 0.427$, Table S1).

The multivariate relationships visualised in the PCA (Figure 3) synthesise these findings, showing that most variability measures are generally positively correlated. The log-variance ratio, acting as a measure of synchrony, was negatively correlated with most other variability metrics (but virtually independent of variability of total biomass, see also Fig. S4), highlighting the significant effect of fertilisation on synchrony ($LVR, f_{1,8} = 6.455$, $p = 0.038$, Table 1), with the synchrony vector pointing towards the unfertilised (NF) plots, particularly to the non-removal ones (NF*NR).

	Factors	CV _{bm}				LVR				CV _{pop}			
		df	F	p		df	F	p		df	F	p	
Between-Plots	Fertil	1	7.456	0,0258	↑	1	2.750	0.136		1	1.358	0.277	
	Removal	1	3.454	0.100		1	0.001	0.977		1	12,299	0,008	↑
	Fertil:Removal	1	0.940	0.361		1	0.731	0.417		1	0	0,998	
	Residuals	8				8				8			
Within-Plots	Detrending	1	2	0.195		1	6,15	0.038	↑	1	74.215	<0,001	↓
	Residuals	8				8				8			
	Factors	CV _{sc}				CV _{sc.euc}				CV _{sc.cho}			
		df	F	p		df	F	p		df	F	p	
Between-Plots	Fertil	1	1,45	0,2636		1	1,19	0,373		1	1,871	0,2085	
	Removal	1	14,5	0,0052	↑	1	13,1	0,0068	↑	1	16,088	0,0039	↑
	Fertil:Removal	1	0,73	0,4194		1	0,36	0,5635		1			
	Residuals	8				8				8			
Within-Plots	Detrending	1	60,2	<0,001	↓	1	31,6	<0,001	↓	1	64,871	<0,001	↓
	Standardis.	1	101	<0,001	↓								
	Residuals	24				8				8			

Table 1: Summary of ANOVA results for stability drivers. Results of the nested analysis of variance (ANOVA) testing the effects of experimental treatments and methodological choices on stability metrics. Bold values indicate statistical significance at $p < 0.05$. Abbreviations: CV_{bm} = species biomass; CV_{sc} = species composition; CV_{sc.euc} = species composition using Euclidean distance; CV_{sc.cho} = species composition using chord distance; LVR = log-variance ratio; CV_{pop} = population variability. The 'Treatment' factor includes Fertilisation (Fertil) and Dominant Removal (Removal), and their interaction (Fertil: Removal). Degrees of freedom (df), F-statistics (F), and p-values (p) are reported for each model component.

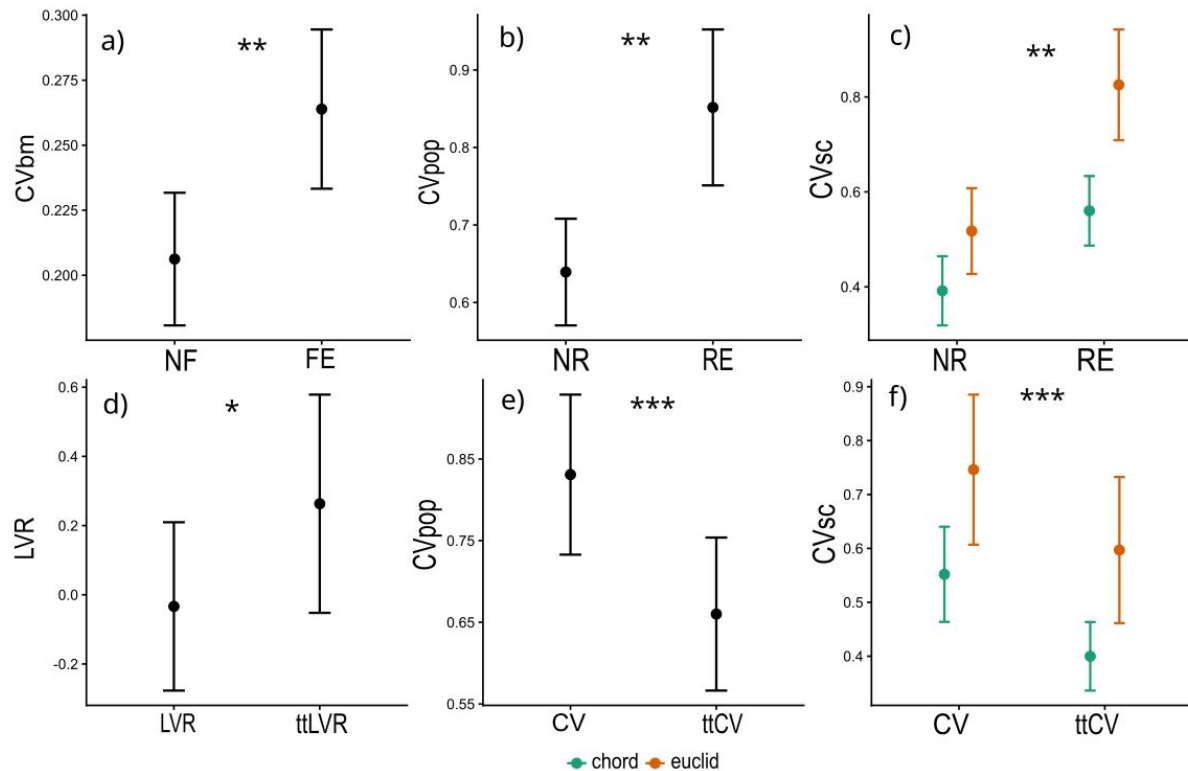


Figure 2: Impact of fertilisation and dominant removal, distance metrics, and detrending on stability components. Responses of biomass variability (CV_{bm}), population variability (CV_{pop}), compositional variability (CV_{sc}), and asynchrony (LVR) to experimental treatments and methodological approaches. Top row (a–c): Effects of fertilisation and dominant removal on standard stability metrics. Boxplots compare unfertilised (NF), fertilised (FE), non-removal (NR) and removal (RE) of dominant plots. For compositional variability (c), the results are further partitioned by the distance metric used: Euclidean distance (representing raw abundance changes in orange) and chord distance (representing standardised relative abundance changes in green). Bottom row (d–f): Comparison between standard and detrended (tt) versions of the metrics. Panels show the log-variance ratio (d), population variability (e), and compositional variability (f). In panel f, the two-distance metrics are distinguished by colour (e.g., orange for Euclidean and green for chord distance). Error bars represent 95% confidence intervals (CIs) calculated directly from the data distribution at the group level. Note that these CIs are descriptive and do not account for the random effects or the pooled variance assumptions used in the ANOVA models; therefore, significant differences indicated by asterisks may occur even where intervals overlap. Significant differences between fertilisation treatments (NF vs. FE) or between standard and detrended metrics are indicated by asterisks (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns = non-significant).

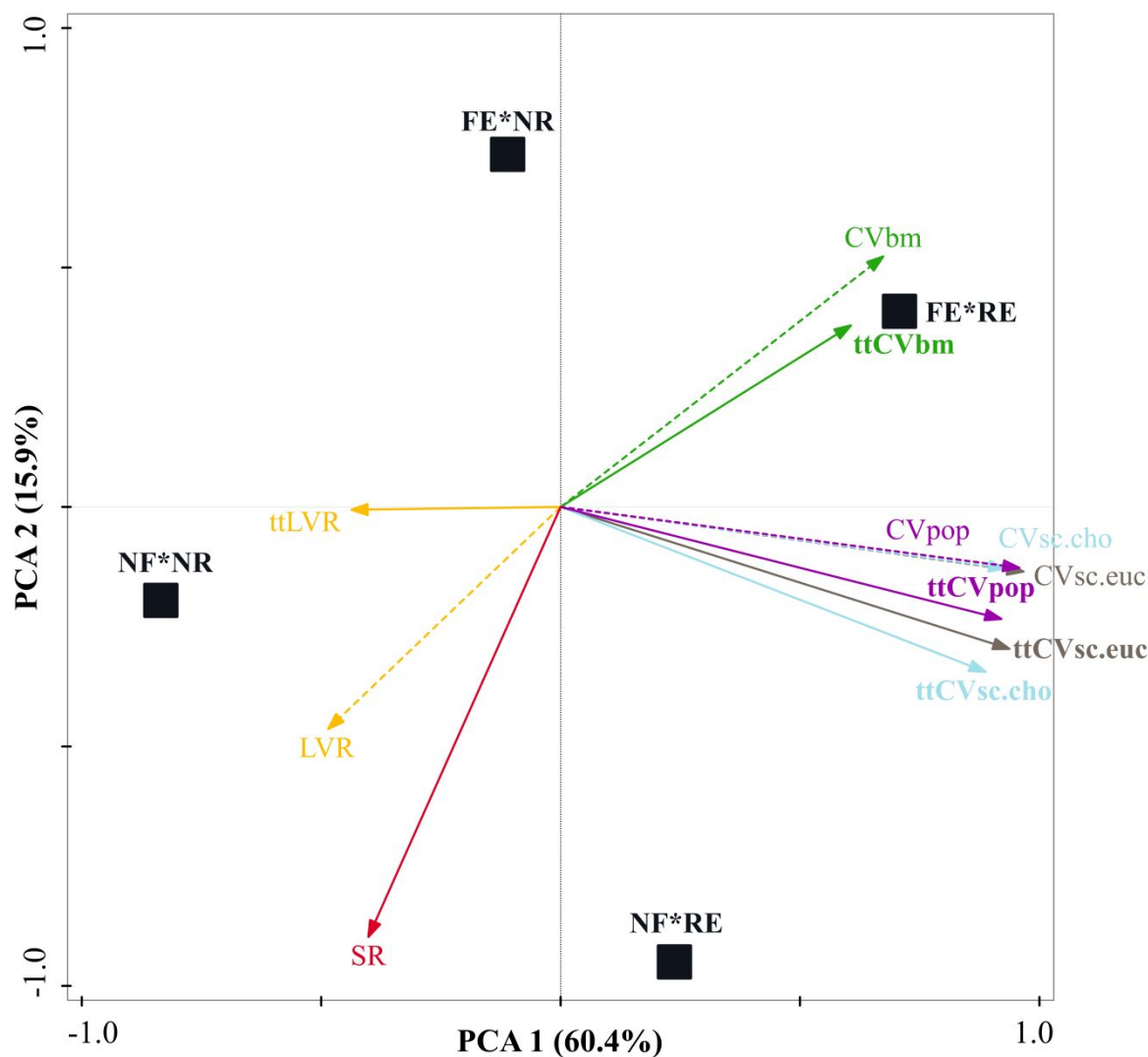


Figure 3: A Principal Component Analysis (PCA) was performed on stability facets and species richness (the average number of species per plot in each year), with interactions between the treatments fertilisation (FE) and removal (RE) as supplementary variables. Explained variability by individual PCA axes in brackets; supplementary variables accounted for 51.97% (adjusted to 41.27%) of the total variability. The PCA biplot shows the relationships between the different stability indices and the experimental treatments. The vectors (arrows) represent the stability drivers, with the length of each vector indicating its contribution to the principal components and the angle between vectors representing their correlation. Abbreviations: CV_{bm} = species biomass; $ttCV_{bm}$ = detrended species biomass, SR = species richness, $CV_{sc.euc}$ = species composition using Euclidean distance, $ttCV_{sc.euc}$ = detrended species composition using Euclidean distance, $CV_{sc.ch}$ = species composition using chord distance, $ttCV_{sc.ch}$ = detrended species composition using chord distance, LVR = log-variance ratio;

ttLVR = detrended log-variance ratio, CV_{pop} = population variability, $ttCV_{pop}$ = detrended population variability.

350

351

4 | DISCUSSION

This study shows that the development and implementation of a battery of detrending metrics for temporal data is essential to provide a more comprehensive picture of the patterns and year-to-year mechanisms of stability. We applied a broad set of detrending indices, including a newly developed one on the stability in species composition over time, to a species-rich grassland under different managements, providing a reinterpretation of the mechanisms of stability already discussed previously (Lepš et al., 2018). The study overall suggests a great potential to apply these indices more routinely and extensively. Specifically, our main finding, that failing to detrend temporal data could potentially lead to overestimation of the importance of compensatory dynamics by asynchrony driven by species turnover and an underestimation of the stabilising contributions of dominant species, should definitely be tested on a broader scale.

Interestingly, detrending had a small effect on the stability of total biomass, as a proxy of productivity, which suggests that variability in biomass is governed by year-to-year fluctuations rather than long-term trends. Ecologically, this is intuitive: biomass production broadly reflects changing weather conditions preceding the biomass assessment that need not be autocorrelated among subsequent years (Knapp & Smith, 2001). Contrastingly, detrending had a strong effect on composition- and population-based metrics (Figure 2 and Table 1). This suggests that the observed compositional changes throughout the duration of the study were driven by long-term directional trends, which we expect can be partially attributable to the sinking water table (Lepš & Lisner 2025). This finding is relevant because it reinforces an earlier concern that stability metrics without regard to temporal order or year-to-year change might distort the effect of long-term directional trends and autocorrelation on synchrony (Lepš et al., 2019; Segrestin et al., 2024). As a result, there is also a risk of misinterpreting the mechanisms contributing to stability. By applying two-term local quadrat variance (TTLQV) and its analogues (Hill, 1973; Lepš et al., 2019), we explicitly separated year-to-year fluctuations from long-term directional change, showing that much of the apparent compositional instability arises from slow trends rather than year-to-year variability.

The average of population CV is a well-established driver of stability (Thibaut & Connolly, 2013; Segrestin et al., 2024), particularly connected to the stability of individual populations. If the populations that form a community are temporally stable, the community itself tends to exhibit more stability. This effect becomes particularly noticeable when

population variability is weighted by species biomass, due to dominant species disproportionately influencing community-level stability (de Bello et al., 2021; Segrestin et al., 2024). The increase in CV_{pop} once the detrending is applied, therefore, indicates that a substantial portion of population variability is caused by long-term trends. A similar pattern emerges for compensatory dynamics among species. Detrending consistently increased synchrony and therefore decreases the effects of compensatory dynamics to species asynchrony. This has already been reported previously (Valencia et al., 2020a; de Bello et al., 2020) and reflects that long-term directional changes can inflate apparent year-to-year species replacements when temporal order is disregarded.

These results overall have important implications for how community stability is described, quantified, and interpreted in classical biodiversity-ecosystem-functioning-experiments. Failing to disentangle long-term directional trends can lead to a systematic lack of understanding of the mechanisms behind stability. Particularly, using non-detrended data from a time series can lead to an overestimation of the role of year-to-year asynchrony, while the stabilising role of dominants, population-level variability will be generally underestimated. Therefore, the distinction between interannual fluctuations and directional trends is critical when linking empirical data to ecological theory, due to the different drivers of community stability operating on different temporal scales.

By introducing a multivariate analogue of the coefficient of variation for species composition, in this paper, we also extend classical CV-based stability concepts to multivariate community data and provide additional insights into the nature of compositional stability and the effects of detrending (Lisner et al., 2024; Lepš & Lisner, 2025). The variability, when calculated through Euclidean distance, shows changes in absolute species biomass; it is inherently linked to fluctuations in total biomass. Using chord distance, however, isolates the changes in relative species proportions. It results in lower variability estimates, which indicates that much of the observed change reflects coordinated changes in absolute abundance (alongside with changes in total biomass across years) rather than reordering of dominance. This is a possible explanation for the positive correlation between biomass and compositional variability (Figure 4), while proportional compositional variability (index) remained relatively constrained.

In line with what was found in previous studies, fertilisation and the removal of dominants consistently destabilised all components of stability. Untreated control plots

exhibited a lower variability of total biomass, species biomass and species composition, and therefore a higher temporal stability (Hautier et al., 2014, 2020; Lepš et al., 2019; Segrestin et al., 2024). Furthermore, fertilisation increased the variability of total biomass, whereas removal of *Molinia caerulea* had strong effects on composition and population level stability (Lepš et al., 2019; Segrestin et al., 2024). Importantly, the variability of total biomass and the variability of species composition were positively correlated. This suggests that fluctuations in biomass and community structure share external drivers, and biomass fluctuations are not independent of community structure changes. The addition of fertiliser shifts the limitation from nutrients to limitation from light, thereby accelerating asymmetric competition. Removal of the dominant *M. caerulea*, on the other hand, disrupts existing community structure and population dynamics by eliminating the strong stabilising influence this species normally has (Lepš et al., 2019; Segrestin et al., 2024).

As shown, fertilisation had a significant effect on asynchrony, increasing it among species (Valencia et al., 2020b). In regard to classical stability theory, this increased asynchrony should stabilise total biomass through compensatory dynamics. However, in our case, this expectation was not met: Fertilised communities were markedly less stable, and while asynchrony should promote stability, the overall stability still declined. This might be a result of the effect of asynchrony not being strong enough to offset the destabilising effects of fertilisation, including the reduction of species richness and an increased population-level variability. The destabilisation observed under fertilisation, however, did not arise from a breakdown of dominants. Instead, it was driven by concurrent reductions in species richness and increases in population-level variability, both of which negatively affect the ability of a community to compensate. Consistent with this dynamic, asynchrony was negatively correlated with biomass stability. Thus, although fertilisation promoted asynchrony, this mechanism alone was insufficient to counteract the destabilising effects of reduced species richness. We did not include an explicit test for the interaction between detrending and treatment presented in Table 1 due to the low number of plots. Instead, we tested the effect of treatments in both detrended and non-detrended data to show that treatment effects were largely consistent across the approaches, with only minor differences which showed that the treatment effects were rather consistent on non-detrended and detrended indices, although future studies, with greater sample size, can nonetheless ideally test to which extent global change drivers can modify stability components while accounting or not for directional trends

REFERENCES

- Byrne, K. M., Lauenroth, W. K., Adler, P. B., & Byrne, C. M. (2011). Estimating Aboveground Net Primary Production in Grasslands: A Comparison of Nondestructive Methods. *Rangeland Ecology & Management*, 64(5), 498–505. <https://doi.org/10.2111/REM-D-10-00145.1>
- de Bello, F., Lavorel, S., Hallett, L. M., Valencia, E., Garnier, E., Roscher, C., Conti, L., Galland, T., Goberna, M., Májeková, M., Montesinos-Navarro, A., Pausas, J. G., Verdú, M., E-Vojtkó, A., Götzenberger, L., & Lepš, J. (2021). Functional trait effects on ecosystem stability: Assembling the jigsaw puzzle. *Trends in Ecology & Evolution*, 36(9), 822–836. <https://doi.org/10.1016/j.tree.2021.05.001>
- Gracia, C., Segrestin, J., Lepš, J., Adler, P. B., Harrison, S. P., Götzenberger, L., Valencia, E., Lecegui, V., Bazzichetto, M., Canullo, R., Chelli, S., Dengler, J., Dolezal, J., Eldridge, D. J., Essl, F., Fischer, F. M., Guido, A., Hallet, L., Herben, T., ... Bello, F. de. (n.d.). A globally consistent scaling relationship reveals stabilizing effects of dominant species in plant communities. *Ecography*, n/a(n/a), e08242. <https://doi.org/10.1002/ecog.08242>
- Hautier, Y., Seabloom, E. W., Borer, E. T., Adler, P. B., Harpole, W. S., Hillebrand, H., Lind, E. M., MacDougall, A. S., Stevens, C. J., & Bakker, J. D. (2014). Eutrophication weakens stabilizing effects of diversity in natural grasslands. *Nature*, 508(7497), 521–525. <https://doi.org/10.1038/nature13014>
- Hautier, Y., Zhang, P., Loreau, M., Wilcox, K. R., Seabloom, E. W., Borer, E. T., Byrnes, J. E. K., Koerner, S. E., Komatsu, K. J., Lefcheck, J. S., Hector, A., Adler, P. B., Alberti, J., Arnillas, C. A., Bakker, J. D., Brudvig, L. A., Bugalho, M. N., Cadotte, M., Caldeira, M. C., ... Wang, S. (2020). General destabilizing effects of eutrophication on grassland productivity at multiple spatial scales. *Nature Communications*, 2020(11), 1–9. <https://doi.org/10.1038/s41467-020-19252-4>
- Hefty, K. L., Povak, N. A., Manley, P. N., Flake, S. W., & Zeller, K. A. (2025). Spatiotemporal patterns of diversity and diversity-stability relationships as a function of compounding disturbances and forest management. *Landscape Ecology*, 40(5), 1–19. <https://doi.org/10.1007/s10980-025-02075-3>
- Hill, M. O. (1973). Diversity and evenness: A unifying notation and its consequences. *Ecology*, 54(2), 427–432. <https://doi.org/10.2307/1934352>
- Jackson, S. T., & Blois, J. L. (2015). Community ecology in a changing environment: Perspectives from the Quaternary. *Proceedings of the National Academy of Sciences of the United States of America*, 112(16), 4915–4921. <https://doi.org/10.1073/pnas.1403664111>

- Knapp, A. K., & Smith, M. D. (2001). Variation among biomes in temporal dynamics of aboveground primary production. *Science*, 291(5503), 481–484. <https://doi.org/10.1126/science.291.5503.481>
- Lepš, J. (1999). Nutrient status, disturbance and competition: An experimental test of relationships in a wet meadow. *Journal of Vegetation Science*, 10(2), 219–230.
- Lepš, J. (2004). Variability in population and community biomass in a grassland community affected by environmental productivity and diversity. *Oikos*, 107(1), 64–71. <https://doi.org/10.1111/j.0030-1299.2004.13023.x>
- Lepš, J. (2014). Scale- and time-dependent effects of fertilization, mowing and dominant removal on a grassland community during a 15-year experiment. *Journal of Applied Ecology*, 51, 978–987. <https://doi.org/10.1111/1365-2664.12255>
- Lepš, J., Götzenberger, L., Valencia, E., & de Bello, F. (2019). Accounting for long-term directional trends on year-to-year synchrony in species fluctuations. *Ecography*, 42(10), 1728–1741. <https://doi.org/10.1111/ecog.04528>
- Lepš, J., & Lisner, A. (2025). Resistance and Resilience of Species Composition: Thirty Years of Experimental Mismanagement and Subsequent Restoration in a Species Rich Meadow. *Ecology and Evolution*, 15(e70923), 1–20. <https://doi.org/10.1002/ece3.70923>
- Lepš, J., Májeková, M., Vítová, A., Doležal, J., & de Bello, F. (2018). Stabilizing effects in temporal fluctuations: Management, traits, and species richness in high-diversity communities. *Ecology*, 99(2), 360–371. <https://doi.org/10.1002/ecy.2065>
- Lisner, A., Segrestin, J., Konečná, M., Blažek, P., Janíková, E., Applová, M., Švancárová, T., & Lepš, J. (2024). Why are plant communities stable? Disentangling the role of dominance, asynchrony and averaging effect following realistic species loss scenario. *Journal of Ecology*, 31, 711–721. <https://doi.org/10.1111/1365-2745.14364>
- Loreau, M., & de Mazancourt, C. (2008). Species synchrony and its drivers: Neutral and nonneutral community dynamics in fluctuating environments. *The American Naturalist*, 172(2), E48–E66. <https://doi.org/10.1086/589746>
- Meteoblue. (2025). *Climate data for Ohrazení, Czech Republic*. (n.d.). Meteoblue. Retrieved 3 November 2025, from https://www.meteoblue.com/cs/climate-change/ohrazen%c3%ad_%c4%8cesko_3069123
- Oksanen, J., Simpson, G. L., Blanchet, F. G., Kindt, R., Legendre, P., Minchin, P. R., O'Hara, R. B., Solymos, P., Stevens, M. H. H., Szoecs, E., Wagner, H., Barbour, M., Bedward, M., Bolker, B., Borcard, D.,

- 509 Borman, T., Carvalho, G., Chirico, M., Caceres, M. D., ... Weedon, J. (2025). *Vegan: Community*
 510 *Ecology Package* (Version 2.7-2) [Computer software]. [https://cran.r-](https://cran.r-project.org/web/packages/vegan/index.html)
 511 [project.org/web/packages/vegan/index.html](https://cran.r-project.org/web/packages/vegan/index.html)
- 512 Orlóci, L. (1967). Data centering, reduction and transformation in ordination. *Journal of Ecology*, 55(1), 205–
 513 223. <https://doi.org/10.2307/2258601>
- 514 Pimm, S. L. (1984). The complexity and stability of ecosystems. *Nature*, 307, 321–326.
 515 <https://doi.org/10.1038/307321a0>
- 516 R Core Team. (2023). *R: A language and environment for statistical computing*. R Foundation for Statistical
 517 *Computing, Vienna, Austria*, <https://www.R-project.org> [Computer software].
- 518 Ravnjak, B., Bavcon, J., & Čarni, A. (2022). Plant Species Turnover on Forest Gaps after Natural Disturbances
 519 in the Dinaric Fir Beech Forests (*Omphalodo-Fagetum sylvaticae*). *Diversity*, 14(3), 209.
 520 <https://doi.org/10.3390/d14030209>
- 521 Schluter, D. (1984). A variance test for detecting species associations, with some example applications.
 522 *Ecology*, 65(3), 998–1005. <https://doi.org/10.2307/1938071>
- 523 Segrestin, J., Götzenberger, L., Valencia, E., Bello, F. de, & Lepš, J. (2024). A unified framework for
 524 partitioning the drivers of stability of ecological communities. *Global Ecology and Biogeography*,
 525 33(5), 1–13. <https://doi.org/10.1111/geb.13828>
- 526 Šmilauer, P., & Lepš, J. (2014). *Multivariate analysis of ecological data using CANOCO 5* (2nd ed.).
 527 Cambridge University Press.
- 528 Ter Braak, C. J. F., & Šmilauer, P. (2012). Canoco 5: Software for multivariate data exploration, testing and
 529 summarization. *Biometrics*.
- 530 Thibaut, L. M., & Connolly, S. R. (2013). Understanding diversity–stability relationships: Towards a unified
 531 model of portfolio effects. *Ecology Letters*, 16(2), 140–150. <https://doi.org/10.1111/ele.12019>
- 532 Valencia, E., Bello, F. de, Lepš, J., Galland, T., E-Vojtkó, A., Conti, L., Danihelka, J., Dengler, J., Eldridge, D.
 533 J., Estiarte, M., García-González, R., Garnier, E., Gómez, D., Harrison, S., Herben, T., Ibáñez, R.,
 534 Jentsch, A., Juergens, N., Kertész, M., ... Götzenberger, L. (2020). Directional trends in species
 535 composition over time can lead to a widespread overemphasis of year-to-year asynchrony. *Journal of*
 536 *Vegetation Science*, 31(5), 792–802. <https://doi.org/10.1111/jvs.12916>
- 537 Valencia, E., De Bello, F., Galland, T., Adler, P. B., Lepš, J., E-Vojtkó, A., Klink, R. van, Carmona, C. P.,
 538 Danihelka, J., Dengler, J., Eldridge, D. J., Estiarte, M., García-González, R., Garnier, E.,

539 Gómez-García, D., Harrison, S. P., Herben, T., Ibáñez, R., Jentsch, A., ... Götzenberger, L. (2020).
540 Synchrony matters more than species richness in plant community stability at a global scale.
541 *Proceedings of the National Academy of Sciences*, 117(39), 24345–24351.
542 <https://doi.org/10.1073/pnas.1920405117>
543 Wang, Y., Wen, S., Farnon Ellwood, M. D., Miller, A. D., & Chu, C. (2018). Temporal effects of disturbance on
544 community composition in simulated stage-structured plant communities. *Ecology and Evolution*, 8(1),
545 120–127. <https://doi.org/10.1002/ece3.3660>
546 Wickham, H. (2015). dplyr: A grammar of data manipulation. *R Package Version 04.*, 3, p156.
547 Wickham, H. (2016). Data analysis. In *Ggplot2: Elegant graphics for data analysis* (pp. 189–201). Springer.
548 Wickham, H., François, R., Bryan, J., & Müller, K. (2023). *dplyr: A Grammar of Data Manipulation* (Version
549 1.1.2.) [Computer software]. <https://cran.r-project.org/web/packages/dplyr/index.html>
550 Wickham, H., Vaughan, D., & Girlich, M. (2023). *tidyr: Tidy Messy Data* (Version 1.3.0.) [Computer software].
551 <https://cran.r-project.org/web/packages/tidyr/index.html>
552

1 **Supplementary material to:**

2 **Title:** Detrending temporal data affects stability drivers but not productivity stability in a
3 species-rich grassland

4 **Authors:**

5 **Summary of contents:** This document provides additional information on the experimental
6 site and the detailed methodological calculations used to derive the metrics. It also contains
7 supplementary figures and tables that support the main results presented in the article.

8

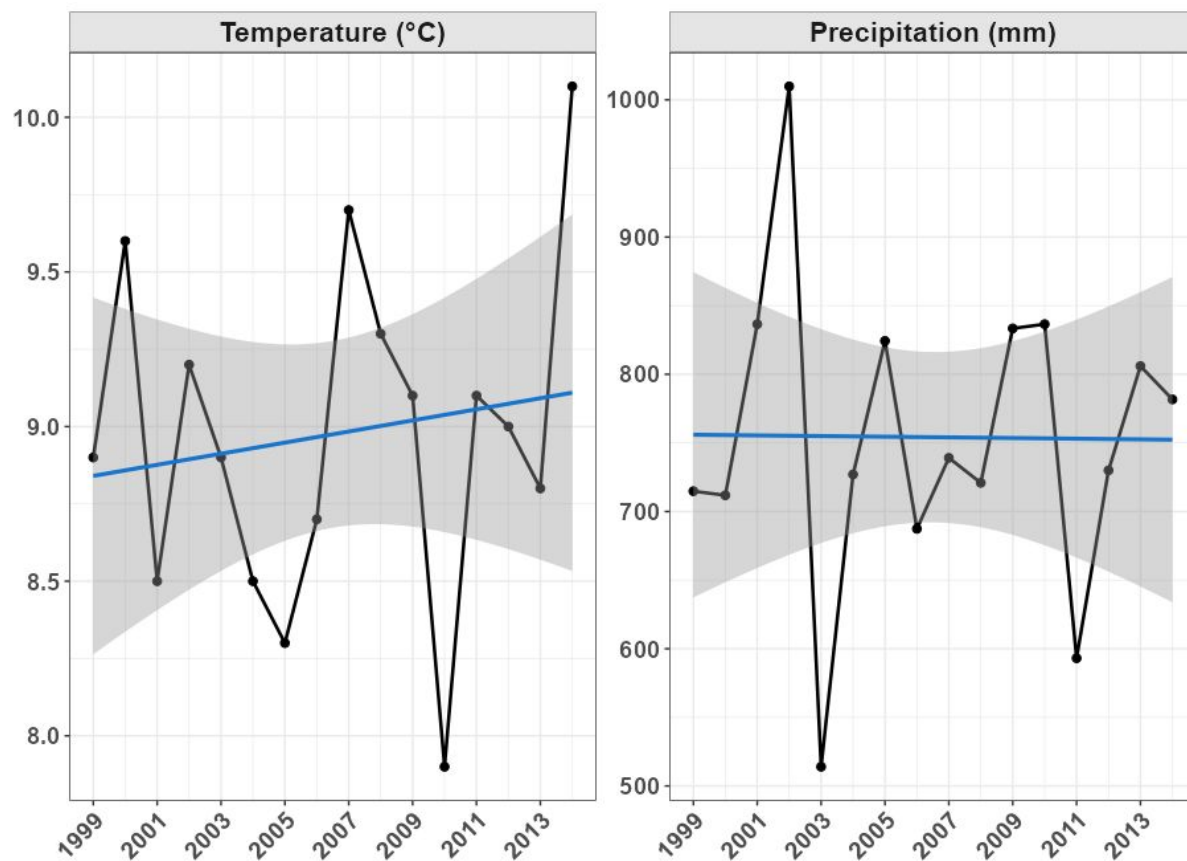
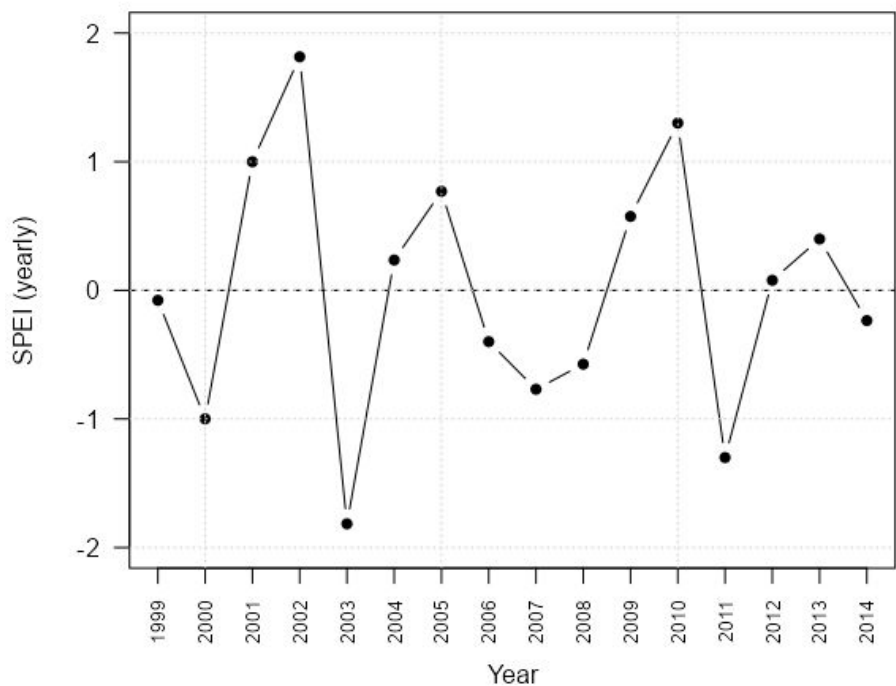


Figure S1a: Changes in mean annual temperature and mean annual precipitation from 1999 to 2014 in Ohrazení, South Bohemia, Czech Republic (48°57'0" N, 14°36'0" E; 510 m above sea level). Linear regressions indicate no significant change in temperature ($p = 0.566$) or precipitation ($p = 0.970$) over time. However, analysis based on a longer time series (1990-2023) shows a significant warming trend (Lepš & Lisner, 2025). Although there was no significant trend in precipitation, the water table decreased over time.

9



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11 Figure S1b: Standardised Precipitation–Evapotranspiration Index (SPEI) values for the study
12 site in Ohrazení over the study period. This time series shows interannual variability with
13 positive values indicating wetter than average conditions and negative values indicating
14 drought conditions. There are several pronounced periods of drought or extreme precipitation,
15 but no consistent directional trend can be seen.

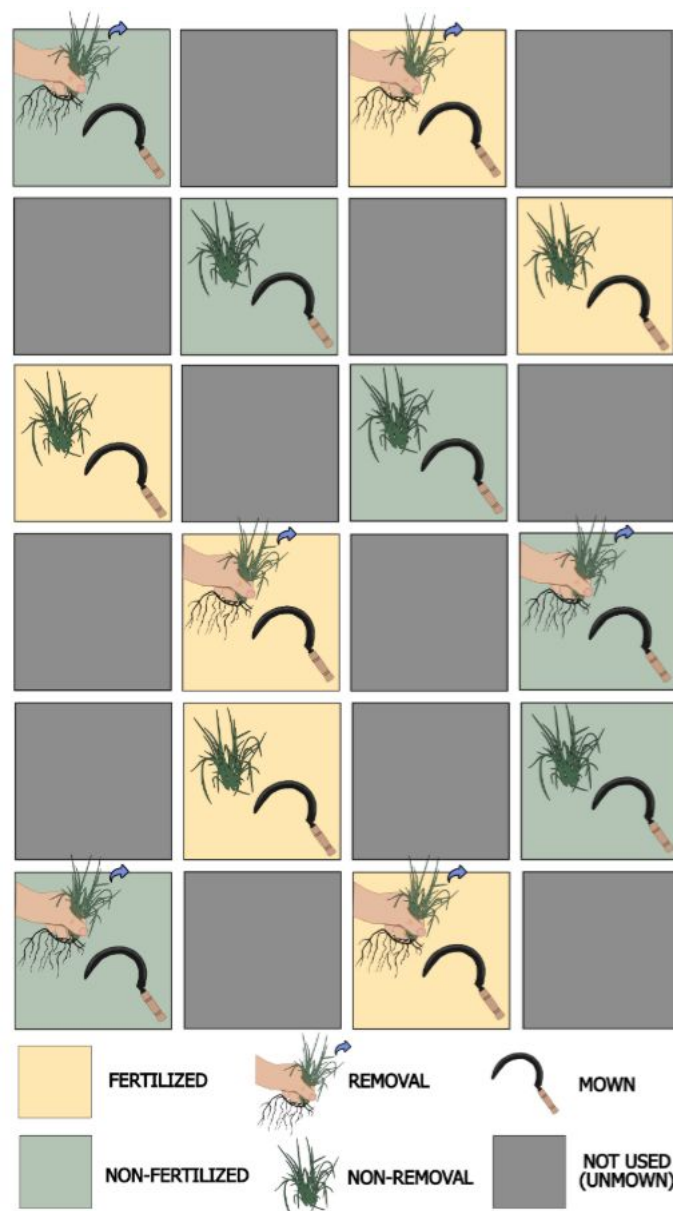


Figure S2: Experimental design and layout of the experiment. Schematic representation of the experimental design at the Ohrazení study site and spatial layout of the experimental blocks. The study follows a randomised complete block design with three replicates. Squares represent the 24 individual 2x2 m plots; of these, the 12 plots shaded in grey were not mown and were therefore excluded from the present analysis. Among the remaining plots, green indicates unfertilized controls, while yellow indicates fertilised plots. The three symbols defined in the integrated legend distinguish between plots where the dominant species (*Molinia caerulea*) was removed, those where it remained, and the management regime, noting that all analysed plots (green and yellow) were subject to regular mowing.



Figure S3: The biomass sampling campaign in the Ohrazeni field entailed the direct identification of taxa in situ (illustrated on the left). This objective was realised through the utilisation of a clipping and sorting process on the samples (illustrated on the right). Subsequently, the remaining vegetation was harvested and sorted in the laboratory.

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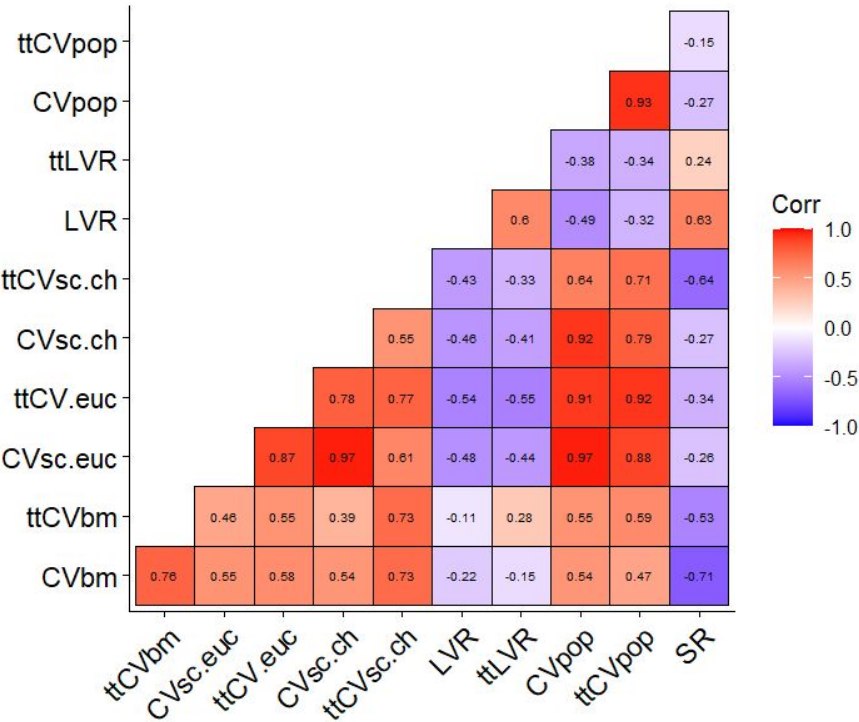


Figure S4: Correlation matrix of ecosystem stability facets and their drivers. Correlogram showing Pearson's correlation coefficients (r) between the different stability and variability metrics calculated for the plant community. The colour scale represents the strength and direction of the correlations, ranging from dark red (strong positive correlation, $r = 1$) to dark blue (strong negative correlation, $r = -1$). White or light-coloured squares indicate weak or non-significant correlations. Abbreviations: CVbm = species biomass; ttCVbm = detrended species biomass, SR = species richness, CVsc.euc = species composition using Euclidean distance, ttCVsc.euc = detrended species composition using Euclidean distance, CVsc.ch = species composition using chord distance, ttCVsc.ch = detrended species composition using chord distance, LVR = log-variance ratio; ttLVR = detrended log-variance ratio, CVpop = population coefficient of variation, ttCVpop = detrended population coefficient of variation. The matrix highlights the decoupling between biomass stability and compositional stability, as well as the strong internal correlations between standard and detrended versions of the same indices.

	CV(bm)				ttCV(bm)			
Treatment	Df	F	p		Df	F	p	
fertil	1	10,336	0,012	↑	1	3,970	0,082	
removal	1	3,579	0,095		1	2,378	0,162	
fertil:removal	1	0,009	0,925		1	1,858	0,210	
residuals	8				8			
	LVR				ttLVR			
Treatment	Df	F	p		Df	F	p	
fertil	1	6,455	0,035	↑	1	0,701	0,427	
removal	1	0,123	0,735		1	0,031	0,864	
fertil:removal	1	0,086	0,776		1	1,157	0,314	
residuals	8				8			
	CV(sc.cho)				ttCV(sc.cho)			
Treatment	Df	F	p		Df	F	p	
fertil	1	3,340	0,105		1	0,326	0,584	
removal	1	15,051	0,005	↑	1	11,730	0,009	↓
fertil:removal	1	2,188	0,177		1	0,024	0,880	
residuals	8				8			
	CV(sc.euc)				ttCV(sc.euc)			
Treatment	Df	F	p		Df	F	p	
fertil	1	2,339	0,165		1	0,357	0,567	
removal	1	13,342	0,006	↑	1	10,752	0,011	↓
fertil:removal	1	0,702	0,426		1	0,034	0,858	
residuals	8				8			
	CV(pop)				ttCV(pop)			
Treatment	Df	F	p		Df	F	p	
fertil	1	2,325	0,166		1	0,552	0,479	
removal	1	14,065	0,006	↑	1	8,859	0,018	
fertil:removal	1	0,085	0,778		1	0,064	0,807	
residuals	8				8			

Table S1. Results of separate analysis of variance (ANOVA) testing the effects of fertilisation (Fertil) and dominant species removal (Removal) on community stability and its drivers. For each metric, results are presented for both the classical version (left columns) and the detrended version (denoted by the prefix ‘tt’, right columns). Bold values indicate statistical significance at $p < 0.05$. Abbreviations: CVbm = species biomass; CVsc = species composition; LVR = log-variance ratio; CVpop = population variability. The 'Treatment' factor includes Fertilisation (Fertil) and Dominant Removal (Removal), and their interaction (Fertil: Removal). Degrees of freedom (df), F-statistics (F), and p-values (p) are reported for each model component.